

Prompt Iteration Log

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The Task (from FL-01 / my real work)

I need a short, specific research-interest paragraph I can use in graduate-school or research-internship applications. It must show that I take machine-learning models from experimentation into working, production-facing pipelines that respect real systems and deployment constraints — not just that I “like AI”.

Version 0 — Naive one-line prompt

Prompt

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Write a research interest paragraph for me about AI and machine learning.
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Output

I am deeply passionate about artificial intelligence and machine learning. I am particularly interested in exploring cutting-edge deep learning techniques and their applications across various domains. My goal is to contribute to the advancement of AI systems that can solve real-world problems and make a positive impact on society. I am eager to collaborate with researchers and continue developing my skills in this exciting field.

Note — Generic, buzzword-heavy, could belong to anyone. No evidence of my actual work or the systems side I care about.

Version 1 — Role Assignment

Technique: Role assignment

Prompt

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You are an experienced research mentor who helps undergraduate students write honest, specific research-interest statements. Write a short research-interest paragraph for a final-year Computer Systems Engineering student who has taken ML models from notebooks into production-facing pipelines and cares about deployment constraints and systems.
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Output

As a final-year Computer Systems Engineering student, I am interested in the systems side of machine learning — specifically the work of moving models from experimental notebooks into reliable, production-facing pipelines. I want to understand how algorithmic choices interact with real hardware, latency, and resource limits, and how those constraints should shape model design rather than being treated as an afterthought.

What changed and why — Giving the model a clear role immediately reduced the generic “passionate about AI” language. The output became more grounded.

Version 2 — Context and Motivation

Technique: Context and motivation

Prompt (abbreviated)

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... + Context about internship, vision trade-offs, systems work. Motivation: the intersection of algorithm and deployment reality. Avoid “passionate”, “cutting-edge”, “leverage”.
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Output

I am interested in the systems side of machine learning: how to move models from experimental settings into pipelines that actually run under real constraints of latency, memory, and hardware. During an AI internship I took models from notebooks into production-facing tools; in computer-vision work I measured the concrete trade-offs between detection accuracy and inference speed. Those experiences convinced me that the interesting questions sit at the boundary between algorithm and system...

What changed and why — Explicit context and motivation removed remaining vagueness. The paragraph now contains concrete evidence and a clear research stance.

Version 3 — Few-shot Examples

Technique: Few-shot examples

What changed and why — The few-shot examples strongly shaped tone and structure. The model stopped explaining “what AI is” and started speaking in the same concrete, systems-oriented voice as the examples.

Version 4 — Output Structure

Technique: Output structure

What changed and why — Imposing a clear four-part structure made the paragraph tighter. Every sentence now has a job: interest → evidence → stance → background.

Version 5 — Step Decomposition (Final)

Technique: Step decomposition

Final output

I am interested in the systems side of machine learning: how models behave once they leave experimental notebooks and must operate under real latency, memory, and hardware constraints. In an AI internship I moved models into production-facing pipelines; in computer-vision work I measured the concrete trade-offs between detection accuracy and inference speed; and I have built the surrounding infrastructure (APIs, agents, deployment paths) that makes those models usable. These experiences showed me that many of the important design questions only appear when a model is forced to live inside a larger system. I want to study how deployment constraints can be treated as first-class inputs to both model design and pipeline architecture. My Computer Systems Engineering background is the practical foundation I bring to that question.

What changed and why — Forcing the model to list evidence and state the research question first produced the cleanest version. Almost all remaining generic language disappeared.

Cross-model comparison (Final prompt)

Claude stayed closer to the “plain and honest” instruction, preserved all three concrete evidence points, and followed structure more reliably. **ChatGPT** produced fluent text but occasionally drifted back toward generic academic language and needed stronger negative constraints. For this task Claude required less post-editing. Both models become useful once the prompt is engineered; the difference is consistency under constraints.

Final reusable prompt template

You are an experienced research mentor... Context: [student’s concrete experience] Motivation: [boundary they care about] + style examples + required four-part structure + constraints (no buzzwords) + silent step decomposition. Output only the final paragraph. (Full template is in the accompanying markdown file.)

This log uses a real task from my FL-01 direction and shows five named techniques applied after the naive baseline.